CAMBRIDGE
International Education

Cambridge O Level

CHEMISTRY **5070/21**

Paper 2 Theory **May/June 2024**

MARK SCHEME

Maximum Mark: 80

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the May/June 2024 series for most Cambridge IGCSE, Cambridge International A and AS Level and Cambridge Pre-U components, and some Cambridge O Level components.

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PUBLISHED

Generic Marking Principles

These general marking principles must be applied by all examiners when marking candidate answers. They should be applied alongside the specific content of the mark scheme or generic level descriptions for a question. Each question paper and mark scheme will also comply with these marking principles.

GENERIC MARKING PRINCIPLE 1:

Marks must be awarded in line with:

- the specific content of the mark scheme or the generic level descriptors for the question
- the specific skills defined in the mark scheme or in the generic level descriptors for the question
- the standard of response required by a candidate as exemplified by the standardisation scripts.

GENERIC MARKING PRINCIPLE 2:

Marks awarded are always **whole marks** (not half marks, or other fractions).

GENERIC MARKING PRINCIPLE 3:

Marks must be awarded **positively**:

- marks are awarded for correct/valid answers, as defined in the mark scheme. However, credit is given for valid answers which go beyond the scope of the syllabus and mark scheme, referring to your Team Leader as appropriate
- marks are awarded when candidates clearly demonstrate what they know and can do
- marks are not deducted for errors
- marks are not deducted for omissions
- answers should only be judged on the quality of spelling, punctuation and grammar when these features are specifically assessed by the question as indicated by the mark scheme. The meaning, however, should be unambiguous.

GENERIC MARKING PRINCIPLE 4:

Rules must be applied consistently, e.g. in situations where candidates have not followed instructions or in the application of generic level descriptors.

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GENERIC MARKING PRINCIPLE 5: Marks should be awarded using the full range of marks defined in the mark scheme for the question (however; the use of the full mark range may be limited according to the quality of the candidate responses seen).
GENERIC MARKING PRINCIPLE 6: Marks awarded are based solely on the requirements as defined in the mark scheme. Marks should not be awarded with grade thresholds or grade descriptors in mind.

Science-Specific Marking Principles

1	Examiners should consider the context and scientific use of any keywords when awarding marks. Although keywords may be present, marks should not be awarded if the keywords are used incorrectly.
2	The examiner should not choose between contradictory statements given in the same question part, and credit should not be awarded for any correct statement that is contradicted within the same question part. Wrong science that is irrelevant to the question should be ignored.
3	Although spellings do not have to be correct, spellings of syllabus terms must allow for clear and unambiguous separation from other syllabus terms with which they may be confused (e.g. ethane / ethene, glucagon / glycogen, refraction / reflection).
4	The error carried forward (ecf) principle should be applied, where appropriate. If an incorrect answer is subsequently used in a scientifically correct way, the candidate should be awarded these subsequent marking points. Further guidance will be included in the mark scheme where necessary and any exceptions to this general principle will be noted.
5	‘List rule’ guidance For questions that require ***n*** responses (e.g. State **two** reasons …): • The response should be read as continuous prose, even when numbered answer spaces are provided. • Any response marked *ignore* in the mark scheme should not count towards ***n***. • Incorrect responses should not be awarded credit but will still count towards ***n***. • Read the entire response to check for any responses that contradict those that would otherwise be credited. Credit should **not** be awarded for any responses that are contradicted within the rest of the response. Where two responses contradict one another, this should be treated as a single incorrect response. • Non-contradictory responses after the first ***n*** responses may be ignored even if they include incorrect science.

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6 Calculation specific guidance

Correct answers to calculations should be given full credit even if there is no working or incorrect working, **unless** the question states 'show your working'.

For questions in which the number of significant figures required is not stated, credit should be awarded for correct answers when rounded by the examiner to the number of significant figures given in the mark scheme. This may not apply to measured values.

For answers given in standard form (e.g. $a \times 10^n$) in which the convention of restricting the value of the coefficient (a) to a value between 1 and 10 is not followed, credit may still be awarded if the answer can be converted to the answer given in the mark scheme.

Unless a separate mark is given for a unit, a missing or incorrect unit will normally mean that the final calculation mark is not awarded. Exceptions to this general principle will be noted in the mark scheme.

7 Guidance for chemical equations

Multiples / fractions of coefficients used in chemical equations are acceptable unless stated otherwise in the mark scheme.

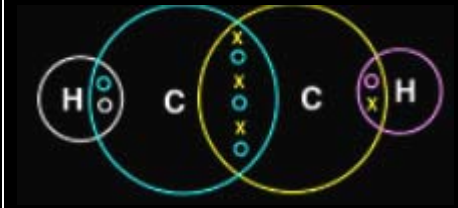
State symbols given in an equation should be ignored unless asked for in the question or stated otherwise in the mark scheme.

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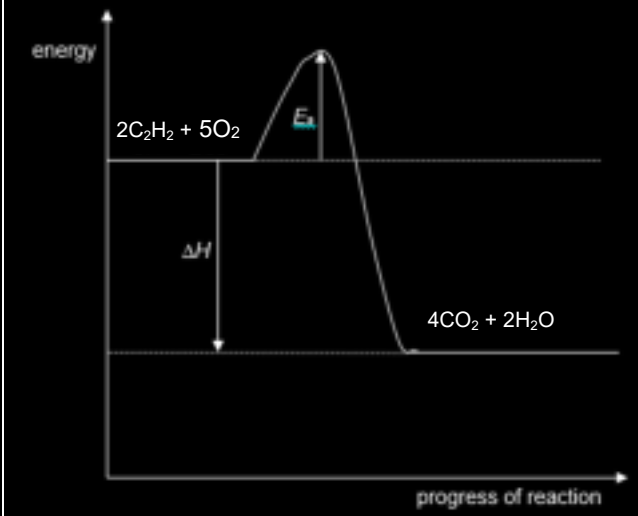
Question	Answer	Marks
1(a)	iron	1
1(b)	silicon(IV) oxide / carbon	1
1(c)	hydrated copper(II) sulfate	1
1(d)	carbon	1
1(e)	glucose	1

Question	Answer	Marks
2(a)	$CaC_2 + 2H_2O \rightarrow Ca(OH)_2 + C_2H_2$ formula for calcium hydroxide as product (1) balanced equation (1)	2
2(b)	C_2^{2-}	1
2(c)(i)	contains only carbon and hydrogen	1
2(c)(ii)	does not contain a carbon-carbon single bond / contains a (carbon-carbon) triple bond	1
2(c)(iii)	orange to colourless	1

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Question	Answer	Marks
2(c)(iv)	six electrons shared between two carbon atoms (1) carbon-hydrogen bonds shown as shared pairs with no lone pairs on either hydrogen or carbon (1) 	**2**
2(d)(i)	bond breaking endothermic **AND** bond making exothermic / energy absorbed to break bonds **AND** energy released on making bonds (1) more energy released than absorbed (1)	**2**

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Question	Answer	Marks
2(d)(ii)	 **M1** products to right of reactants **AND** reactant level above product level (1) **M2** enthalpy change shown as downward arrow **AND** labelled as enthalpy change or ΔH (1) **M3** activation energy drawn to maximum of energy hump with an upward arrow **AND** labelled (1)	**3**

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Question	Answer	Marks
3(a)	amount of ammonium nitrite 0.025 × 0.133 **OR** 0.003325 (mol) **AND** amount of nitrogen = 0.003325 (mol) (1) volume of nitrogen = amount × 24 **OR** 0.0798 (mol) (1) volume of nitrogen = 0.080 (dm^3) (1)	**3**
3(b)	rate increases because particles move faster / particles have more kinetic energy / particles have gained kinetic energy (1) more successful collisions / more collisions/particles) with equal or more than activation energy / more effective collisions / more energetic collisions (1)	**2**
3(c)	rate decreases / reaction is slower because particles are less crowded / distance between particles is smaller / fewer particles per unit volume (1) fewer collisions per second / lower collision frequency (1)	**2**
3(d)	universal indicator (paper) (1) match colour with a pH colour chart (1)	**2**

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Question	Answer	Marks
4(a)	strong attraction between positive and negative ions	**1**
4(b)	two potassium (atoms) lose 1 electron each (1) iodine (molecule) gains two electrons (1)	**2**
4(c)	at anode – iodine / I_2 (1) at cathode – hydrogen / H_2 (1)	**2**
4(d)	purple (1) to brown (1)	**2**
4(e)	I^- or iodide oxidised since it loses electrons (1) Cl_2 or chlorine reduced since it gains electrons (1)	**2**

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Question	Answer	Marks
5(a)	(position of equilibrium) moves to the right / (position of equilibrium) moves to the product side / (position of equilibrium) moves to sulfur dioxide or zinc oxide side (1) to absorb thermal energy (1)	**2**
5(b)	(position of equilibrium) moves to the right / (position of equilibrium) moves to the product side / (position of equilibrium) moves to sulfur dioxide or zinc oxide side (1) more moles of **gas** on right hand side / more moles of **gas** on product side (1)	**2**
5(c)	M_r of $ZnSO_3$ is 145 **AND** of ZnO is 81 **OR** M_r of $ZnSO_3$ is 65 + 32 + 48 **AND** of ZnO is 65 +16 (1) amount of $ZnSO_3$ and ZnO is 25.5 / 145 **OR** 0.1759 (mol) (1) mass of zinc oxide is 14.2448 (g) (1)	**3**
5(d)	zinc oxide is amphoteric (1) sulfur dioxide is acidic (1)	**2**
5(e)	$ZnSO_3(s) + 2HNO_3(aq) \rightarrow Zn(NO_3)_2(aq) + H_2O(l) + SO_2(g)$ balanced equation (1) state symbol dependent on correct formulae (1)	**2**

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Question	Answer	Marks
6(a)	climate change	**1**
6(b)	(carbon dioxide) absorbs thermal energy (from the Earth) (1) reflection of thermal energy (back to the Earth) / emission of thermal energy (towards the Earth) (1) reduces/stops loss of thermal energy into space (1)	**3**
6(c)(i)	carbon dioxide + water → glucose + oxygen	**1**
6(c)(ii)	needs chlorophyll **AND** (to absorb) light energy	**1**
6(d)	any one from: reforestation / afforestation to absorb more carbon dioxide (by photosynthesis) (1) decreasing use of fossil fuels (to stop emitting carbon dioxide) (1) use more renewable energy resources (to reduce carbon dioxide being emitted) / use hydrogen as a fuel (which produces water rather than carbon dioxide) (1)	**1**

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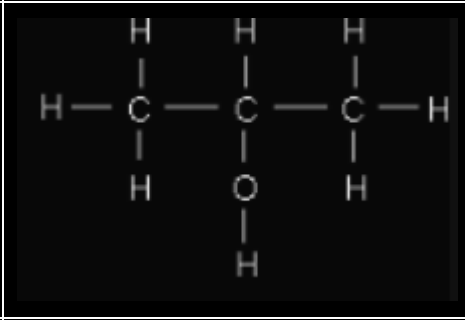
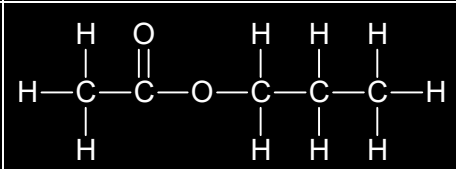
<table>
<tr><th>Question</th><th>Answer</th><th>Marks</th></tr>
<tr><td>7(a)</td><td>volume decreases (1)

particles move closer to one another / distance between particles decreases (1)</td><td>2</td></tr>
<tr><td>7(b)</td><td>particle separation – move apart (1)

arrangement – ordered to random (1)

motion – vibrating to moving from one place to another (1)</td><td>3</td></tr>
<tr><td>7(c)</td><td>chlorine has a lower (relative) molecular mass / chlorine molecules have less mass</td><td>1</td></tr>
<tr><td>7(d)</td><td><table><tr><th>particle</th><th>number of particles</th></tr><tr><td>electrons</td><td>54</td></tr><tr><td>neutrons</td><td>73</td></tr><tr><td>protons</td><td>53</td></tr></table></td><td>3</td></tr>
</table>

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Question	Answer	Marks
8(a)(i)	<u>propan-1-ol</u>	**1**
8(a)(ii)		**1**
8(b)		**1**
8(c)	CH_3CH_2COOH	**1**
8(d)	name – magnesium propanoate (1) formula – $Mg(CH_3CH_2COO)_2$ (1)	**2**
8(e)	**E** – carbon dioxide (1) **F** – hydrogen (1)	**2**
8(f)	<u>ethyl propanoate</u>	**1**

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<table>
<tr><th>Question</th><th>Answer</th><th>Marks</th></tr>
<tr><td>9(a)</td><td>any one comment about addition polymerisation
addition polymerisation (involves monomers joining together with) no other product formed (1)

addition polymer has same empirical formula as the monomer (1)

addition polymers uses monomers with a C = C bond (1)

any one comment about condensation polymerisation
condensation polymerisation involves (monomers joining together with) the formation of a simple molecule (1)

condensation polymer does not have same empirical formula as monomer(s) (1)

condensation polymerisation uses monomers with two functional groups (1)</td><td>2</td></tr>
<tr><td>9(b)</td><td>ester</td><td>1</td></tr>
<tr><td>9(c)</td><td><table>
<tr><td>element</td><td>C</td><td>H</td><td>Cl</td></tr>
<tr><td>%</td><td>47.1</td><td>6.5</td><td>46.4</td></tr>
<tr><td>amount</td><td>47.1 / 12 OR 3.93</td><td>6.5 / 1 OR 6.5</td><td>46.4 / 35.5 OR 1.31</td></tr>
<tr><td>ratio</td><td>3.00 / 3</td><td>4.96 / 5</td><td>1</td></tr>
</table>
amount row (1)

ratio (1)

C_3H_5Cl (1)</td><td>3</td></tr>
</table>

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Question	Answer	Marks
9(d)	**Any two from:** land-fills (may fill up) (1) accumulation of plastics in oceans (1) formation of toxic gases during **burning** (1)	**2**

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